Effects of Epidural Spinal Stimulation on Cortical Activity and Motor Output During Power Grip in Cervical Spinal Cord Injury

Milton O. Candela, JeongHoon Oh, Dimitry G. Sayenko, Alexander G. Steele

Department of Neurosurgery, Center for Neural Systems Restoration, Houston Methodist Hospital, Houston, TX, USA

Background

Spinal Cord Injury (SCI) can disrupt the corticospinal tract (CST), impairing sensory feedback and motor control below the level of injury.

In complete cervical SCI (AIS A), individuals show absent upper-limb EMG and force, and altered M1 activity [1-2] that no longer encodes force effectively [3] due to disrupted internal motor circuits [4] (Fig. 1).

Epidural Spinal Stimulation (ESS) facilitates volitional muscle activation and enhances force generation after an SCI [5], but its effect on M1 force encoding activity remains unclear.

Here, we examined cortical (EEG) and motor (EMG, force) changes during a power grip task in two individuals with complete chronic cervical SCI (AIS A).

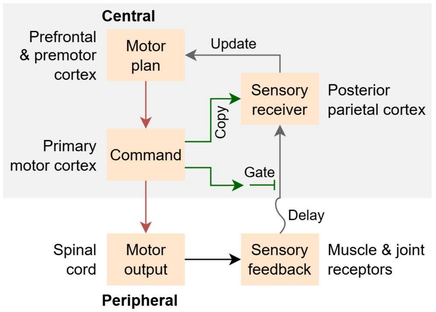


Fig. 1 | During limb movement, motor commands activate motor neurons and muscle contractions, generating proprioceptive feedback. Temporal delays require 1) a rapid internal feedback copy, and 2) gated sensory feedback. Adapted by Grafstein B. from [4].

Experimental Design

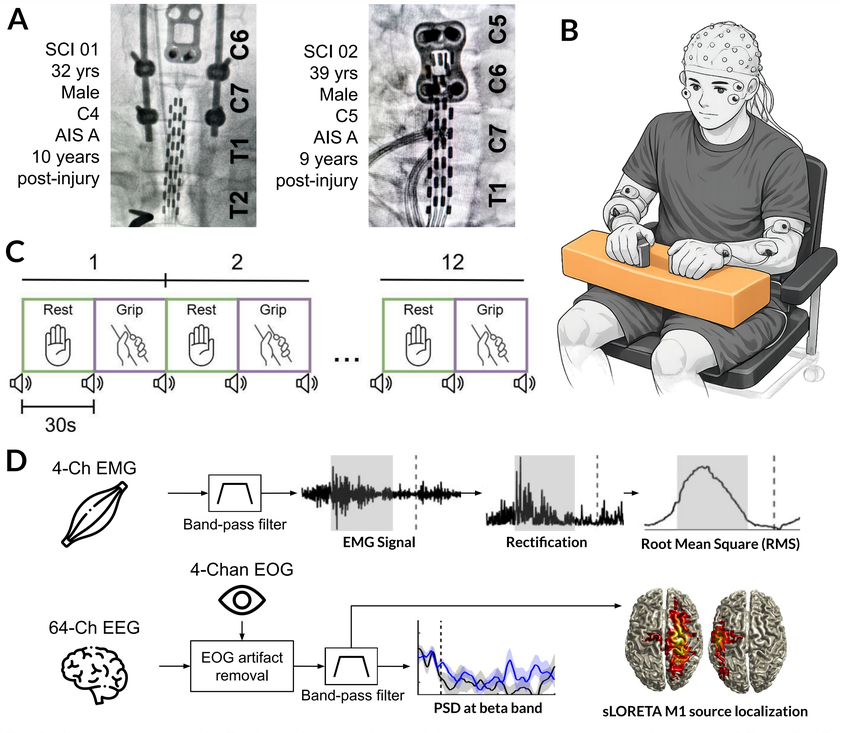


Fig. 2 | Study overview. (A) Clinical trial involving individuals with complete chronic cervical SCI (n=2), **(B)** Schematic of experimental setup, including EEG, EOG, EMG, and force measurements; participants were seated and their hands were supported by a cushion. **(C)** Alternating rest-grip trials (n=12), each consisting of fixed 30s with auditory cues. **(D)** Signal acquisition and processing pipeline for EMG, EEG, and EOG data.

ESS on Motor Output

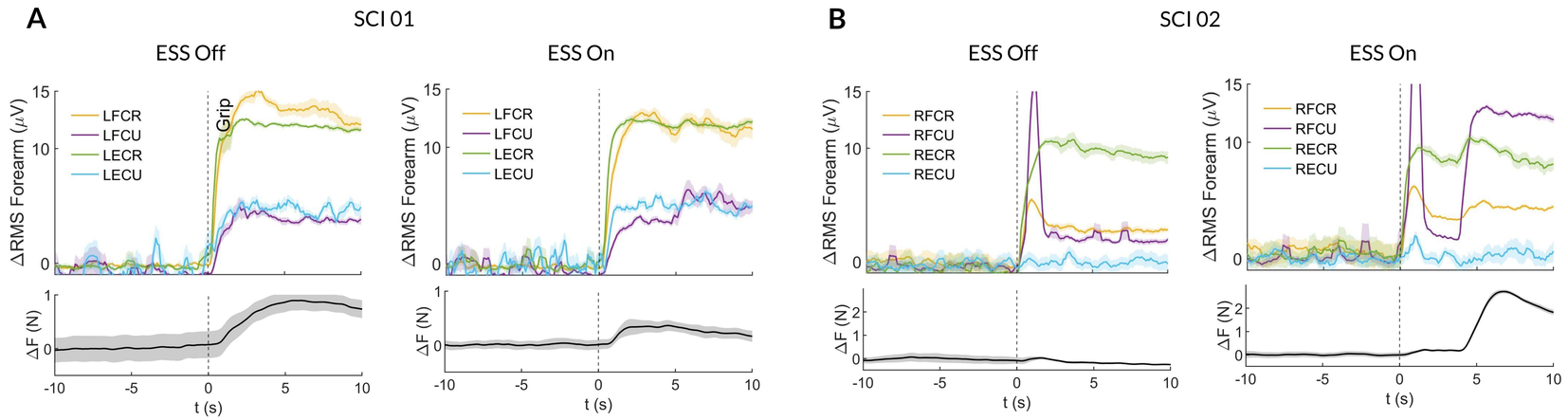


Fig. 3 | ESS changed EMG and force consistently. (A, B) Baseline subtraction [-10, 0] for change in force (ΔF), and change in RMS (ΔRMS) of forearm muscles (Flexor Carpi Radialis [FCR], Flexor Carpi Ulnaris [FCU], Extensor Carpi Radialis [ECR], Extensor Carpi Ulnaris [ECU]).

ESS on Cortical Activity

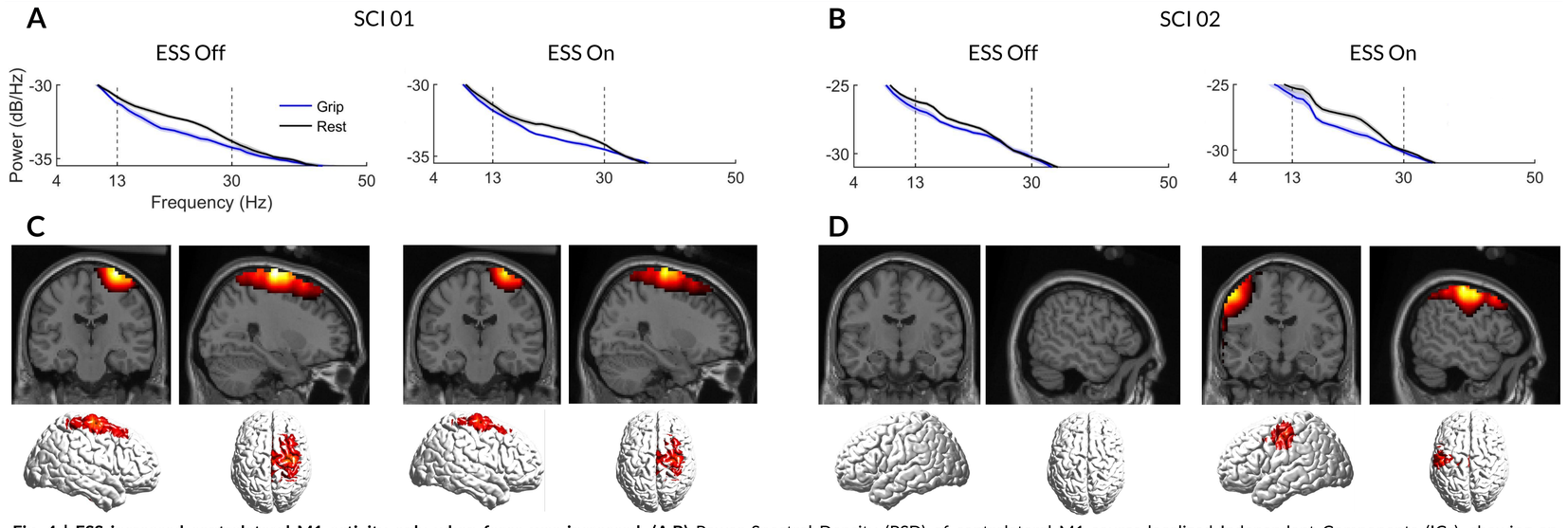


Fig. 4 | ESS increased contralateral M1 activity only when force was increased. (A,B) Power Spectral Density (PSD) of contralateral M1 source-localized Independent Components (ICs), showing an Event-Related Desynchronization (ERD) at the beta (β, 13-30 Hz) rhythm. **(C,D)** sLORETA projection maps of beta PSD effect size (permutation test) when compared to rest (orthogonal and surface views).

Conclusion

- In individuals with chronic cervical complete SCI, sub-lesional ESS can facilitate motor output (Fig. 3) associated with cortical activation (Fig. 4). Consistent with enhanced engagement of residual descending and ascending pathways.
- However, further studies with larger cohorts, alternative neuroimaging techniques, and varied motor tasks are needed to confirm these findings.

References

[1] Cremoux et al., 2013, *Clin. Neurophysiol.* [2] Foldes et al., 2017, *J. Neurophysiol.* [3] Kalaska, 2009, *Prog. Mot. Control* [4] Azim, 2014, *Science* [5] Powell et al., 2023, *Nat Med.*

Acknowledgements

This project was supported by Paula and Joseph C. “Rusty” Walter III, the Walter Oil & Gas Corporation, and the Neurospark seed fund.